

EXPERIENCE

- Aurora Innovation** Senior SWE / Tech Lead (Behavior ML) 2022–present
- Led team of 7 ML engineers to develop & deploy forecasting models, reducing error by 20% on targeted scenarios, resolving 100+ behavioral test failures across 7 safety-critical failure modes, & enabling driverless operations that achieved 100K+ incident-free miles.
 - Recognized by senior leadership for urgently resolving intersection behavior deficiencies blocking 2 investor showcases central to \$800M+ funding round & manufacturing partnership agreement.
 - Mentored 3 ML engineers, & developed framework accelerating integration of separate model improvements from 1 week to 2 days, enabling the team to double in size while maintaining development velocity.
- Magna Electronics (formerly Optimus Ride)** SDE II → Senior SDE (Motion Planning) 2020–2022
- Led 3 engineers to develop probabilistic uncertainty quantification framework & risk-aware motion planner, enabling safe autonomous navigation through contested intersections.
 - Led cross-functional initiative to redesign prediction-planning interface incorporating probabilistic uncertainty propagation, facilitating risk-aware planning across the autonomous driving stack.
 - Formulated technical requirements & algorithmic specifications for 2 ADAS features following ISO 26262 functional safety standards, resulting in 1 patent filing.
- Cruise Automation** Intern (Prediction) Summer 2018
- Improved forecast dynamic feasibility & intersection reasoning, tripling passing test cases & substantially reducing intersection disengagements.
- Optimus Ride (Self-driving)** SWE Intern (Perception) Summer 2017
- Designed & implemented end-to-end perception pipeline from object detection through tracking & forecasting, establishing architecture that integrated with motion planning system.
- RightHand Robotics (Robotic Grasping)** Intern (Machine Learning / Vision) January 2017
- Designed & trained CNN-based grasp prediction model from RGB-D input, improving pick success rate on cluttered bin-packing scenarios.
- Tanius Technology (Prop Trading)** Developer (ML Modeling) 2015–2016
- Built end-to-end ML system for algorithmic trading: data ingestion, model training, & backtesting infrastructure.

EDUCATION & RESEARCH PUBLICATIONS

- Massachusetts Institute of Technology | CSAIL Learning & Intelligent Systems** 2016–2020
- M.Eng** Electrical Engineering & Computer Science (AI Concentration), GPA 5.0 (out of 5.0)
- S.B.** Computer Science + Brain & Cognitive Science, GPA 4.9 (out of 5.0), 5.0 within CS major
- B. Axelrod*, **L. Shimanuki***, "Efficient Motion Planning under Obstacle Uncertainty with Local Dependencies" WAFR 2022
- L. Shimanuki***, B. Axelrod*, "Hardness of Motion Planning with Obstacle Uncertainty in 2 Dimensions" IJRR 2021
- B. Kim*, **L. Shimanuki***, et al., "Representation, learning, and planning algorithms for geometric task and motion planning" IJRR 2021
- B. Kim*, **L. Shimanuki***, "Learning value functions with relational state representations for guiding task-and-motion planning" CoRL 2019
- L. Shimanuki***, B. Axelrod*, "Hardness of Motion Planning with Obstacle Uncertainty in 3 Dimensions" WAFR 2018

* indicates equal contribution

SKILLS Languages: C, C++, Python ML/AI: PyTorch, TensorFlow, ROS Tools: Git, Docker, Linux

ACTIVITIES

Site Manager	<i>Food for Free COVID-19 Relief Program</i>	2020
Directed team of 12 volunteers for packing & delivering groceries to ~300 households weekly.		
Program Director, Head Webmaster	<i>MIT Educational Studies Program</i>	2017–2020
Directed educational programs reaching ~3000 MS/HS students, with ~1000 classes taught by ~500 teachers, & run by ~100 volunteers. Mentored 5 future program directors. Maintained website used by ~5000 students.		
Software Lead	<i>AVBotz</i>	2012–2016
Led the software team (~12 members) for fully-autonomous submarine for manipulating objects, aiming & shooting torpedoes, & navigating around obstacles. International finalist (7th Place) at RoboSub 2015.		
Co-President, HPMS Branch Director	<i>ACE Coding</i>	2013–2016
Managed ~16 volunteers to teach weekly programming lessons to ~100 middle school students annually. Organized full-day event attracting ~300 students. Taught machine vision & ML workshops.		

AWARDS

USACO Platinum	Intel STS 2016 Semifinalist	Eagle Scout	MIT Battlecode 2018 Finalist
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PROJECTS

C++	Low latency audio streaming to enable remote musicians to play in-sync with under 30ms latency
Python	Musical autocomplete to assist chord & melody composition
Python	Open-source window manager extension enabling grid layout
C	Web browser using Chromium's rendering engine with configurable vi-like key bindings
C	C compiler to convert C code to x86 assembly
Java	Neural network AI for a multiplayer platformer fighting game
n/a	Inflatable Sailboat: DIY spritsail rigged to an inflatable raft